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PRN: B26ET2024 Div: SY3 Class: C3

INPUT:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
#include <math.h>
#define SIZE 80

struct Stack {
   double data[SIZE];
   int top;
} st;

void push(double val) {
   if (st.top >= SIZE - 1) {
      printf("\nError: Stack Overflow\n");
   } else {
      st.top++;
      st.data[st.top] = val;
   }
}

double pop() {
   if (st.top == -1) {
      printf("\nError: Stack Underflow\n");
      return 0;
   } else {
      double val = st.data[st.top];
      st.top--;
      return val;
   }
}

double evaluate_postfix(char exp[]) {
   int i = 0;
   double op1, op2;
```

```
    st.top = -1;

    while (exp[i] != '$' && exp[i] != '\0') {
        if (isdigit(exp[i])) {
            push((double)(exp[i] - '0'));
        } else {
            op2 = pop();
            op1 = pop();

            switch (exp[i]) {
                case '+': push(op1 + op2); break;
                case '-': push(op1 - op2); break;
                case '*': push(op1 * op2); break;
                case '/':
                    if (op2 == 0) {
                        printf("\nError: Division by zero\n");
                        return 0;
                    }
                    push(op1 / op2);
                    break;
                case '^': push(pow(op1, op2)); break;
                default:
                    printf("\nError: Invalid character '%c'\n", exp[i]);
                    return 0;
            }
        }
        i++;
    }
    return pop();
}
int main() {
    char exp[SIZE];
    int len;

    printf("Enter a postfix expression (e.g., 234*+): ");
    if (scanf("%78s", exp) != 1) {
        printf("Error reading input.\n");
        return 1;
    }
```

```
    len = strlen(exp);
    exp[len] = '$';
    exp[len + 1] = '\0';

    double result = evaluate_postfix(exp);
    printf("The value of the expression is: %g\n", result);

    return 0;
}
```

OUTPUT:

Output

```
Enter a postfix expression (e.g., 234*+): 54+362/*-
The value of the expression is: 0
```